INCH-POUND
MIL-DTL-365B
26 April 2023
SUPERSEDING
MIL-Z-365A
18 March 1968

DETAIL SPECIFICATION

ZINC-DUST
(FOR USE IN PYROTECHNICS)

Reinstated after 26 April 2023 and may be used for new and existing designs and acquisitions.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one grade of zinc-dust for use in pyrotechnics.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

COMMERCIAL ITEM DESCRIPTIONS

A-A-59282 - Chemicals, Analytical; General Specification for

(Copies of this document are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE FORMS

DD Form 250 - Material Inspection and Receiving Report

(Copies of this document are available online at www.esd.whs.mil.)

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 6810

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OPNAV INSTRUCTIONS

OPNAVINST 4440.27 - Storage and Handling of Liquefied and Gaseous Compressed Gases and Their Full and Empty Cylinders

(Copies of this document are available online at www.secnav.navy.mil/doni/default.aspx.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY FOR QUALITY (ASQ)

ANSI/ASQ Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at www.asq.org.)

ASTM INTERNATIONAL

ASTM E11 - Standard Specification for Woven Wire Test Sieve Cloth and Test Sieves

ASTM E323 - Standard Specification for Perforated-Plate Sieves for Testing Purposes

(Copies of these documents are available online at www.astm.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Material. The zinc-dust shall meet the chemical and granulation requirements specified herein. In addition, the zinc-dust shall be uniform in quality and free from foreign material. Documents such as DD Form 250 can be a valuable tool in providing material and order traceability.

3.2 Chemical requirements. The zinc-dust shall conform to the applicable chemical requirements specified in table I.

TABLE I. Chemical requirements.

Property	Percentage		Test Paragraph
	Maximum	Minimum	
Total zinc, calculated as zinc	—	97.5	4.3.2
Metallic zinc	—	94.0	4.3.3
Impurities (other than zinc oxide)	2.0	—	4.3.3.2.2
Zinc oxide	6.0	—	4.3.4
Moisture and other volatile matter	0.1	—	4.3.5
Matter soluble in organic solvent mixture	0.0	—	4.3.6

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3.3 Granulation. The zinc-dust shall conform to the granulation requirements of table II when determined in accordance with 4.3.7.

TABLE II. Granulation requirements.

Sieve number [1/]	Percent through minimum
100	99
200	90
230	75

NOTE:

[1/] Use U.S. Standard sieves conforming to ASTM E11 and ASTM E323.

4. VERIFICATION

4.1. Conformance inspection. Each zinc-dust sample obtained per 4.2.2 shall conform to the requirements of this specification per the tests and examination of 4.3. Unless otherwise specified (see 6.2), all chemicals shall be American Chemical Society (ACS) grade in accordance with A-A-59282, and distilled water shall be used. OPNAVINST 4440.27 may be consulted for proper storage and handling of compressed gases and compressed gas containers.

4.2 Sampling.

4.2.1 Lot size. For the purpose of sampling, a lot of zinc-dust shall consist of a manufacturer's batch (see 6.3). If the material cannot be identified by batch, a lot shall consist of not more than 1,000 pounds (lb) of zinc-dust offered for delivery at one time (see 6.2).

4.2.2 Sample for tests. From each lot offered to the Government, two 1 lb samples of zinc-dust shall be removed from separate unit containers taken at random.

4.2.3 Sample for examination of filled containers. A random sample of filled containers shall be selected from each lot of zinc-dust offered to the Government in accordance with ANSI/ASQ Z1.4 at inspection level I and the acceptance quality limit (AQL) specified (see 6.2).

4.2.4 Rejection criteria. If a sample fails to meet any of the test requirements of this specification, the lot represented by the sample shall be rejected.

4.3 Test methods.

4.3.1 Visual inspection. Conformance of the zinc-dust to the requirements for material (see 3.1) shall be determined by visual inspection.

4.3.2 Total zinc. Total zinc shall be determined by the potassium ferrocyanide method.

4.3.2.1 Standardization of potassium ferrocyanide solution. For standardization of potassium ferrocyanide solution, the following procedure shall be used:

a. Weigh 0.2 gram (g) of metallic zinc or freshly ignited ZnO. Then transfer it to a 400-milliliter (mL) beaker, and dissolve in 10 mL of HCl (specific gravity 1.18) and 20 mL of water.

b. Drop a small piece of litmus paper in the 400-mL beaker, add NH_4OH until slightly alkaline, add HCl until just acid, and then 3 mL more of HCl.

c. Dilute the mixture to about 250 mL with hot water and heat to nearly boiling.

d. Add in $K_4Fe(CN)_6$ solution while stirring slowly and constantly until a drop of the solution tested with a drop of uranyl nitrate indicator on a white porcelain plate exhibits a brown tinge after standing for 1 minute.

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e. The $K_4Fe(CN)_6$ solution is prepared by dissolving 22 g of $K_4Fe(CN)_6 \cdot 3H_2O$ in water and diluting it to 1 liter.

f. The uranyl-nitrate indicator is prepared by dissolving 5 g of $UO_2(NO_3)_2 \cdot 6H_2O$ in 100 mL of water. During the titration, the temperature of the solution shall not be allowed to fall below 70 °C.

g. A blank determination shall be made using the same amounts of reagents and water, under the same conditions of temperature, volume, and acidity obtained when the standardized sample is titrated.

4.3.2.1.1 Calculation. The strength of the $K_4Fe(CN)_6$ standard solution in terms of grams of zinc shall be calculated by the following formula:

$$Zinc\ equivalent\ in\ grams\ per\ milliliter\ of\ K_4Fe(CN)_6\ solution = \frac{W}{T_1 - T_2}$$

Where:

W = weight of zinc sample (or equivalent to the ZnO used)

T_1 = mL of $K_4Fe(CN)_6$ used in titration of sample

T_2 = mL of $K_4Fe(CN)_6$ used in titration of blank

4.3.2.2 Procedure. The following procedure shall be used:

a. Weigh 0.25 g of the powdered zinc, transfer it to a 400-mL beaker, moisten with alcohol, and dissolve in 10 mL of HCl (specific gravity 1.18) and 20 mL of water.

b. Continue with the procedure used in 4.3.2.1, beginning with the addition of litmus paper and adjustment of the acidity with NH_4OH and HCl (see 4.3.2.1.b).

4.3.2.3 Calculation. The percentage of total zinc shall be calculated by the following formula:

$$Percentage\ of\ total\ zinc = \frac{(X - X_1)B\ x\ 100}{W}$$

Where:

X = mL of $K_4Fe(CN)_6$ used in titration of sample

X_1 = mL of $K_4Fe(CN)_6$ used in titration of blank

B = zinc equivalent in grams per milliliter (g/mL) of $K_4Fe(CN)_6$ solution (see 4.3.2.1.1)

W= weight of zinc sample (g)

4.3.3 Metallic zinc. Metallic zinc shall be determined by the hydrogen evolution method.

4.3.3.1 Apparatus. For the apparatus, the following procedure shall be used:

a. Assemble the apparatus as shown on figure 1.

b. Calibrate bulb D to include capacity from stopcock E to the zero reading of the burette.

c. Also calibrate the capacity of flask A and tubes to stopcocks E and B.

d. Saturate the water in leveling bulb F and measuring tube D with hydrogen at room temperature and pressure.

4.3.3.2 Procedure.

4.3.3.2.1 Total hydrogen evolved. For total hydrogen evolved, the following procedure shall be used:

a. Weigh 1.15 g of the powdered zinc and transfer it to flask A.

b. Also add to flask A: approximately 5 g pure ferrous ammonium sulfate; a piece of sheet platinum approximately 5 square centimeters; and an amount of water exactly 100 mL less than the capacity of the flask and tubes.

c. Connect the apparatus as show on figure 1.

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d. Open stopcocks B and E, raise F to bring the water in the burette to stopcock E, and then close F and B. This leaves 100 mL of air in the apparatus.

e. Read the temperature and barometric pressure, and calculate the gas volume to standard temperature and pressure by the following formula:

$$V = \frac{Pv}{760} \, x \frac{1}{1 + 0.00367t}$$

Where:

V = corrected volume (mL)

v = original volume (mL)

P = barometer reading in millimeters (mm) corrected for the vapor pressure of water at t (°C)

Note: In order to evaluate the vapor pressure of water, since flask A contains acid and the measuring tube contains water, it may be assumed that the entire static system is over water.

t = temperature in degrees Celsius

0.00367 = coefficient of expansion of hydrogen and air for each degree rise in temperature.

760 = normal atmospheric pressure

f. Now pour 43 mL of 1:1 sulfuric acid into the thistle tube, lower F, open stopcock E, and allow the acid to run into flask A, partially opening stopcock B. During the admission of the acid, avoid air admission.

g. Close stopcock B and allow the apparatus to stand until the gas volume becomes constant, keeping the level of liquid in F and D fairly equal throughout the evolution to avoid leakage and hydrogen solubility changes due to pressure differences. The evolution is complete in about 3 hours.

h. Calculate the total volume of gas in the apparatus by adding 60 mL to the gas measuring apparatus reading. These 60 mL are the volume of the gas remaining in flask A after the addition of 43 mL of acid. (The 43 mL of 1:1 sulfuric acid adds 40 mL to the liquid due to shrinkage of mixing.)

i. Read the temperature and the barometric pressure and correct the total gas volume by the preceding formula for deviations from standard temperature (0 °C) and pressure (760 mm) (see 4.3.3.2.2.1.e).

j. Subtract the corrected volume of gas originally in the apparatus from the corrected volume.

k. Add 2 mL to the remainder to correct for the solubility of hydrogen in the solution in flask A. The sum is the hydrogen evolved at standard temperature and pressure.

4.3.3.2.2 Impurities. For accurate results, the volume of hydrogen evolved (see 4.3.3.2.1) shall be corrected for the hydrogen evolved by iron, which is usually present, and for tin, which may be present. To make these corrections, the apparatus shown on figure 2 shall be assembled.

4.3.3.2.2.1 Iron. For iron (see figure 2), the following procedure shall be used:

a. Weigh approximately 11.5 g of the sample, and place it in flask B. Then add a piece of sheet platinum approximately 5 square centimeters and close the apparatus tightly.

b. Pass nitrogen through the apparatus until the air is removed.

c. Close pinchcock G and stop the flow of nitrogen.

d. Add 125 mL of water through separatory funnel B to flask B. Then add 50 mL of concentrated sulfuric acid dropwise through the separatory funnel.

e. During the addition of the acid, keep the contents of the flask at or below room temperature by means of an ice bath.

f. After the addition of the acid, allow the flask to stand for the length of time required for the hydrogen evolution in the determination of metallic zinc, which usually requires 3 hours.

g. Open pinchcock G, and draw the contents of flask B through the asbestos mat in tared filter F and into flask B[1] by means of suction applied at H.

h. Wash flask B four times by adding a 50-mL portion of water and shaking and drawing over into flask B[1].

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i. Remove the stopper from flask B[1] and titrate immediately with potassium permanganate solution to a pink color permanent for 1 minute.

j. Calculate the reducing matter as percentage of iron (Fe) by the following formula:

$$Reducing\ matter\ calculated\ as\ percentage\ of\ Fe = \frac{5.584\ AN}{W}$$

Where:

A = mL of $KMnO_4$ used in titration

N = normality of $KMnO_4$ solution

W = weight of zinc-dust sample

4.3.3.2.2.2 Residue. For residue, the following procedure shall be used:

a. Complete the washing of the residue in the tared filter F (see 4.3.3.2.2.1), and dry at 105 to 110 °C.

b. Cool the residue in a desiccator and weigh.

c. Calculate the percentage of residue insoluble in dilute sulfuric acid by the following formula:

$$Percentage\ of\ residue = \frac{100A}{W}$$

Where:

A = weight of dried residue

W = weight of zinc-dust sample

4.3.3.2.2.3 Tin. For tin (Sn) (see figure 2), the following procedure shall be used:

a. Transfer the titrated solution from flask B[1] (see 4.3.3.2.2.1) to a 600-mL beaker.

b. Add NH_4OH until just alkaline, and then add 5 mL of HCl or 5 mL of H_2SO_4 (1:3) for each 100 mL of solution.

c. Cover the beaker, place it on a hot plate, and heat to about 95 °C. Then pass in a slow stream of H_2S for 1 hour, digest for 1 hour, and allow to stand one-half hour longer.

d. Filter and wash the precipitate of SnS alternately with three portions each of wash solution (100 mL of saturated $NH_4C_2H_3O_2$ solution, 50 mL of glacial $HC_2H_3O_2$, and 850 mL of water) and hot water.

e. Transfer the filter paper and precipitate to a 50-mL beaker. Add 10 to 20 mL of ammonium polysulfide, heat to boiling, and filter.

f. Repeat the digestion twice with the ammonium polysulfide and the filtration, and then wash the filter with hot water.

g. Acidify the combined filtrate and washings with $HC_2H_3O_2$ (1:9), digest on a hot plate for 1 hour, allow it to stand overnight, and filter through an ignited, cooled and weighed Gooch crucible with an asbestos pad.

h. Wash alternately with two portions each of the wash solution and hot water, and dry thoroughly.

i. Ignite over a Bunsen flame, very carefully at first to convert the sulfide to the oxide, and then partly cover the crucible and heat strongly over a large Bunsen or Meker burner. (SnS must be converted carefully to the oxide, which may be heated to a high temperature without loss by volatilization.)

j. Weigh as SnO_2 and calculate percentage of metallic tin by the following formula:

$$Percentage\ of\ Sn = \frac{78.77A}{W}$$

Where:

A = weight of dried residue

W = weight of sample

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4.3.3.3 Calculations. The following calculations shall be performed:

a. Multiply the percentage of tin (see 4.3.3.2.2.3.j) by 0.94, and subtract the result from the reducing matter calculated as percentage of iron (see 4.3.3.2.2.1.j). The remainder is percentage of metallic iron.

b. Correct the evolved hydrogen at standard temperature and pressure (see 4.3.3.2.1) for the iron and tin present in the sample taken for the determination of metallic zinc on the basis that 0.01 g metallic iron evolves 4.012 mL of hydrogen, and 0.01 g metallic tin evolves 1.887 mL of hydrogen at standard temperature and pressure.

c. Calculate the volume corrected for the iron and tin present as metallic zinc by the following formula:

$$Percentage\ of\ mettalic\ zinc = \frac{0.29188M}{W}$$

Where:

M = mL of hydrogen corrected to standard temperature and pressure

W = weight of sample

The sum of the percentages of residue, iron, and tin shall equal the percentage of impurities other than zinc oxide.

4.3.4 Zinc oxide. Zinc oxide shall be determined by finding the difference between the percentage of total zinc and percentage of metallic zinc.

4.3.4.1 Calculation. The percentage of zinc oxide shall be calculated by the following formula:

$$Percentage\ of\ zinc\ oxide = (A - B)1.2447$$

Where:

A = percentage of total zinc

B = percentage of metallic zinc

4.3.5 Moisture and other volatile matter. For moisture and other volatile matter, the following procedure shall be used:

a. A weighed portion of approximately 10,000 g of the sample shall be transferred to a tared glass bottle, heated for 2 hours in an oven at 100 to 105 °C, cooled in a desiccator, and weighed.

b. The loss in weight as percentage of moisture shall be calculated as follows:

$$Loss\ in\ weight\ as\ percentage\ of\ moisture = \frac{100(A - B)}{W}$$

Where:

A = weight of glass bottle and sample

B = weight of glass bottle

W = weight of sample

4.3.6 Matter soluble in organic-solvent mixture. For matter soluble in organic-solvent mixture, the following procedure shall be used:

a. Transfer 25 g of zinc-dust to a 250-mL beaker, add 100 mL of dry solvent, stir with a glass rod for 2 minutes, and filter through a Gooch crucible prepared with an asbestos pad into a 200-mL tared beaker.

b. Evaporate the solvent on a steam bath, dry the beaker and contents in an oven at 100 to 105 °C for 1 hour, and cool in a desiccator, and weigh.

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c. When treated as specified, 100 mL of the solvent mixture shall leave no weighable residue.

Organic solvent mixture:

10 volumes ethyl ether

6 volumes benzene

4 volumes methanol (95 to 100 percent)

1 volume acetone

The solvent mixture shall be neutral. If it tests either acid or alkaline, it shall be purified by shaking with pure zinc-dust and decanting therefrom and distilling.

4.3.7 Granulation. For granulation, the following procedure shall be used:

a. Fit three sieves conforming to ASTM E11 and ASTM E323, numbers 100, 200, and 230 U.S. Standard series, together in the order of decreasing mesh size, placing the coarsest mesh on top and a receiving pan at the bottom.

b. Place a weighed portion of 100 g of the zinc-dust sample on the upper sieve, cover the upper sieve, and shake for 5 minutes by means of a mechanical shaker geared to produce 300±15 gyrations and 150±10 taps of the striker per minute.

c. Weigh the quantity retained on each sieve, calculate the percentage of zinc-dust that has passed through each sieve, and check for conformance to table II.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Metallic zinc powder, commercially known as zinc-dust, covered by this specification is intended for use in pyrotechnics as they apply to Loaded GTR – 18A rockets. Existing stock of Rocket, Loaded GTR – 18A manufactured to MIL-Z-365A, dated 18 March 1968, is acceptable for use until depleted.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Quantity required in pounds (see 4.2.1).

c. AQL of 2.5 percent defective (see 4.2.3)

d. Packaging requirements (see 5.1).

6.3 Batch. A batch is defined as that quantity of material that has been subjected to some unit chemical or physical mixing process intended to make the final product substantially uniform.

6.4 Metallic zinc. A rapid method for determining metallic zinc is given in ASTM D521. The results are inclined to be somewhat low. For the highest accuracy, the hydrogen evolution method detailed in 4.3.3 should be used.

6.5 Subject term (key word) listing.

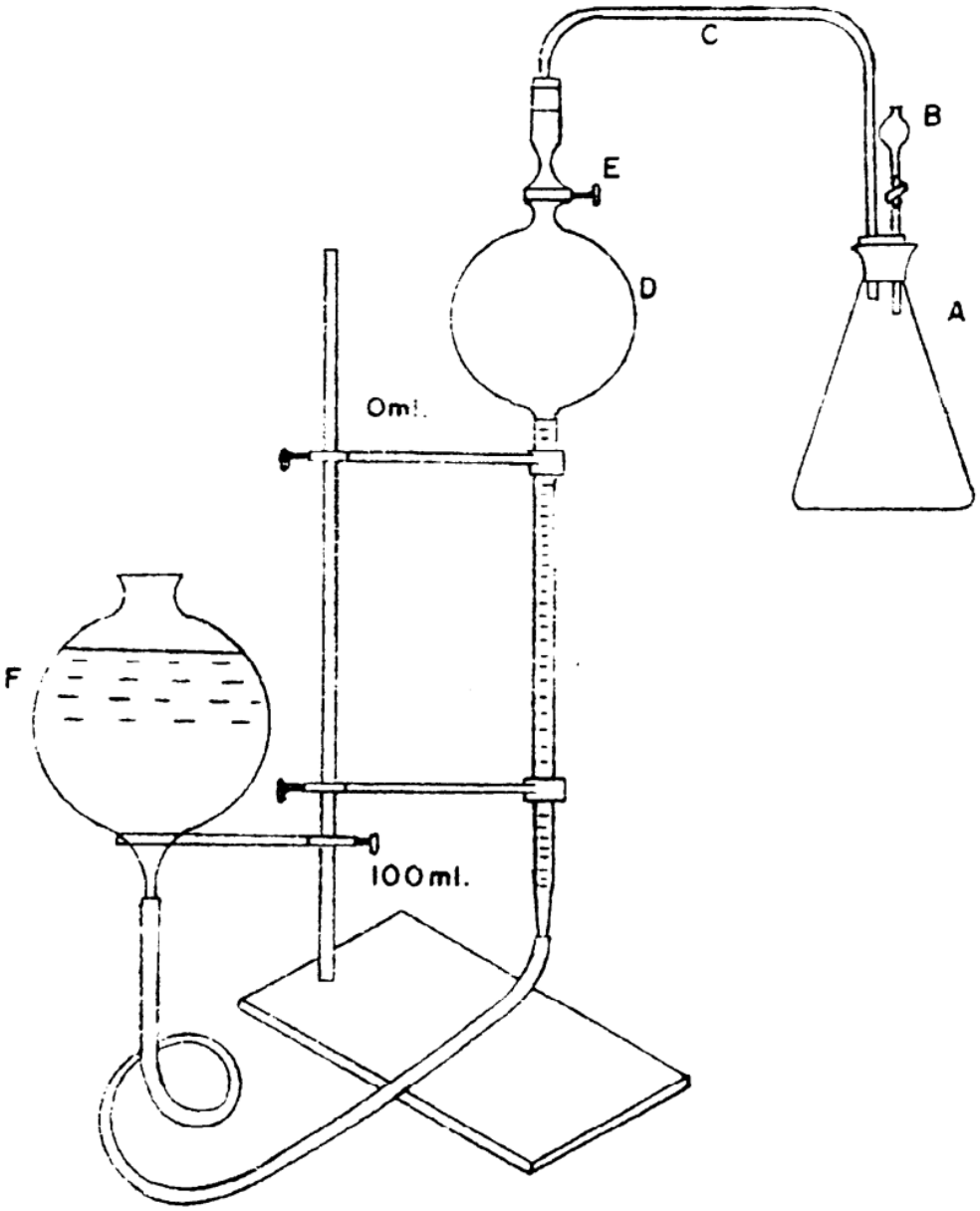
Zinc Powder

Pyrotechnics

Rocket, GTR – 18A

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6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.



A – Flask of 220 mL capacity when filled to stopper

B – Thistle tube with stopcock

C – Glass tubing connection

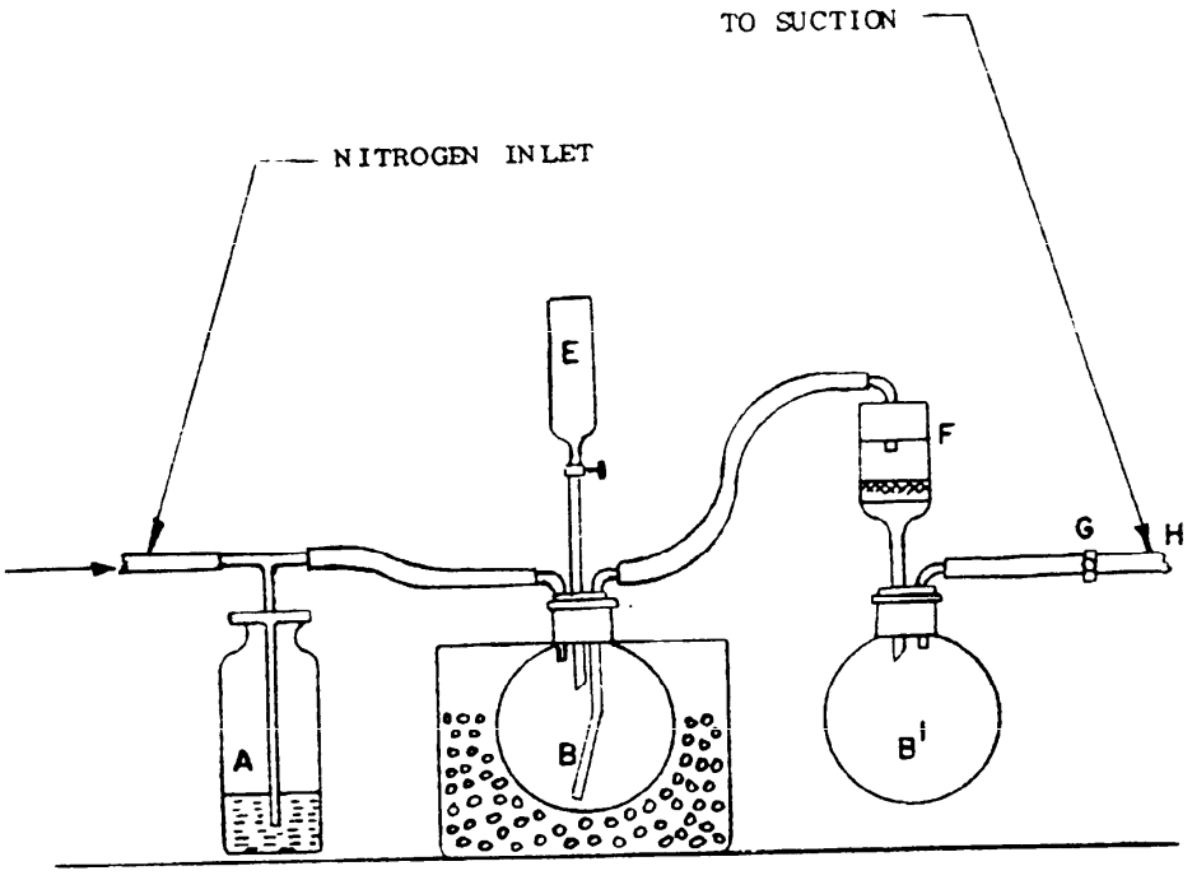
D – Bulb of 400 mL capacity with gas burette graduated from 0 to 100 mL in 0.2 mL subdivision

E – Stopcock for gas burette

F – Leveling bulb

FIGURE 1. Apparatus for the total hydrogen evolved (part of metallic zinc determination).

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A – Wide mouth bottle, 16 ounces, containing 2 inches of water

B and B^1 – Round bottom ring neck flasks, 500 mL

E – Separatory funnel, 60 mL, with 15-centimeter stem

F – Fritted glass filter, 1½ inches in diameter, provided with an asbestos mat

G – Pinchcock

FIGURE 2. Apparatus for iron and tin (part of metallic zinc determination).

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